
CP3AI









Dance Fam!



FIFO <3













ABOUT ME

- Senior studying CS (Minor: Psychology)
- Pleasanton (Bay Area)
- 3rd year in Makers
- Previous Projects: DanceCam + FIFO
- Likes: art, matcha, good food, mangoes,

  GSW, bowling (recent)



:)

MY INSPIRATION?

- Game Pigeon / Ateball
- Figuring out when to remove cups,
  whose turn it is, rearranging cups, etc.
- Best Cup Pong Duo: Kaylee + Hannah

WHAT IS CP3AI?

Cup Pong (3) AI || Chris Paul 3 Allen Iverson



Cup Pong



Cup Pong @arcade

CP3AI is an automated Cup Pong system that combines computer vision, embedded systems, and robotics to track and manage a game in real time. The system will detect successful shots, automatically update the game state, and use a motorized mechanical system to remove and rearrange cups for repositioning. LEDs will provide visual feedback, while a web or mobile interface will allow players to view game information and control the system.

COMPUTER VISION

- Use computer vision to build the shot tracking system
- Detect cups and identify when successful shots occur
- Send detected events to the rest of the systems





GAME SOFTWARE

- Develop the core software that manages gameplay and game state from start to finish
- Integrate shot detection from CV system
- Create web/mobile dashboard for viewing the live game

ELECTRICAL

- Control and manage embedded hardware such as LEDs, motors, and other electronic components for automation
- Translate game-state updates into physical actions
- Use LEDs to communicate shots, cup states, game events







- Design and construct the physical table
- Develop a motorized mechanism to remove and reposition cups
- Design mounts for cameras and electrical components

MECHANICAL



FALL

Demonstrate one functional side of the game:

- Computer vision system correctly tracks successful shots
- Real-time game-state tracking with responsive LED feedback
- Integration between microcontroller, software, and hardware





SPRING

Demonstrate full 2-player game:

- Automated re-racking system
- Bonus: add gameplay analytics/player stats

WHY JOIN MY TEAM ?

1

Fun project :)

2

Roles for every major/discipline + control over design

3

AI is involved in this project which is super relevant rn!

4

Get closer as a team with team bonding! (food + fun activities)







THANK YOU

Contact: Hannah Lee

Email: hlee4786@usc.edu

Phone: (925) 998-5635